MARINE INTERNAL COMBUSTION ENGINES (MME 467)

SUPERCHARGING AND TURBOCHARGING

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Introduction

- ❖ It will be the principal aim of the engine designer, to achieve the twin goals of improved power output and minimum exhaust emissions.
- ❖ The power output of a naturally aspirated engine depends mainly on the following five factors:
  - ✓ Amount of air inducted into the cylinder.
  - ✓ Extent of utilization of the inducted air.
  - ✓ The speed of the engine.
  - ✓ Quantity of fuel admitted and its combustion characteristics.
  - ✓ Thermal efficiency of the engine.
- ❖ The most preferred method of increasing the power output is by means of increasing the mean effective pressure.
  - ✓ This can be achieved by supplying air or air-fuel mixture at a pressure that is higher than the atmospheric pressure.
  - ✓ This will increase the density, thereby the mass of air or air-fuel mixture inducted for the same swept volume. This in turn will increase the power output of the engine.

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Introduction

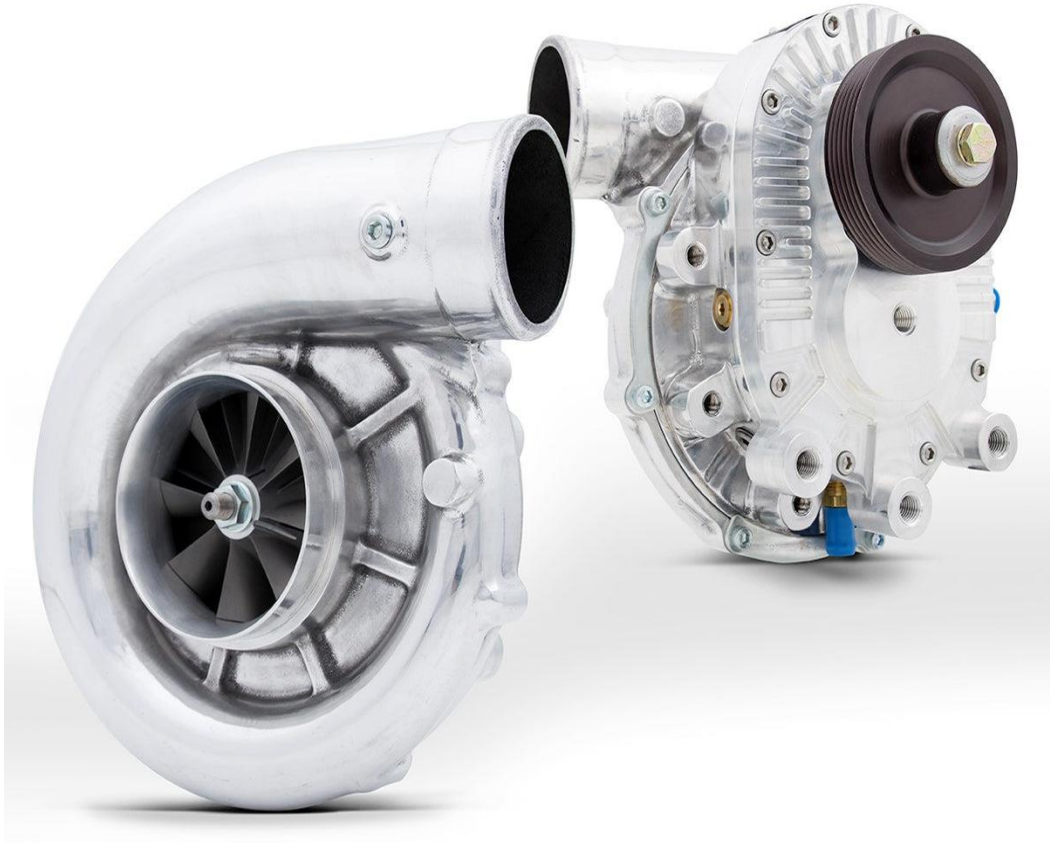
- This method of supplying air or fuel-air mixture higher than the pressure at which the engine naturally aspirates, by means of a boosting device is called supercharging.
- The device which boosts the pressure is called a supercharger.
- Supercharging of internal combustion engines is in practice for a long time as a method for improving engine power output.
- The trend points to small displacement engines in order to meet emission legislation on fuel consumption and emission control.
- The consumers, however, still demand equally the same or better performance they are used to.
- ✓ A good way to meet these needs is to have supercharging which may be called forced induction.





Supercharging

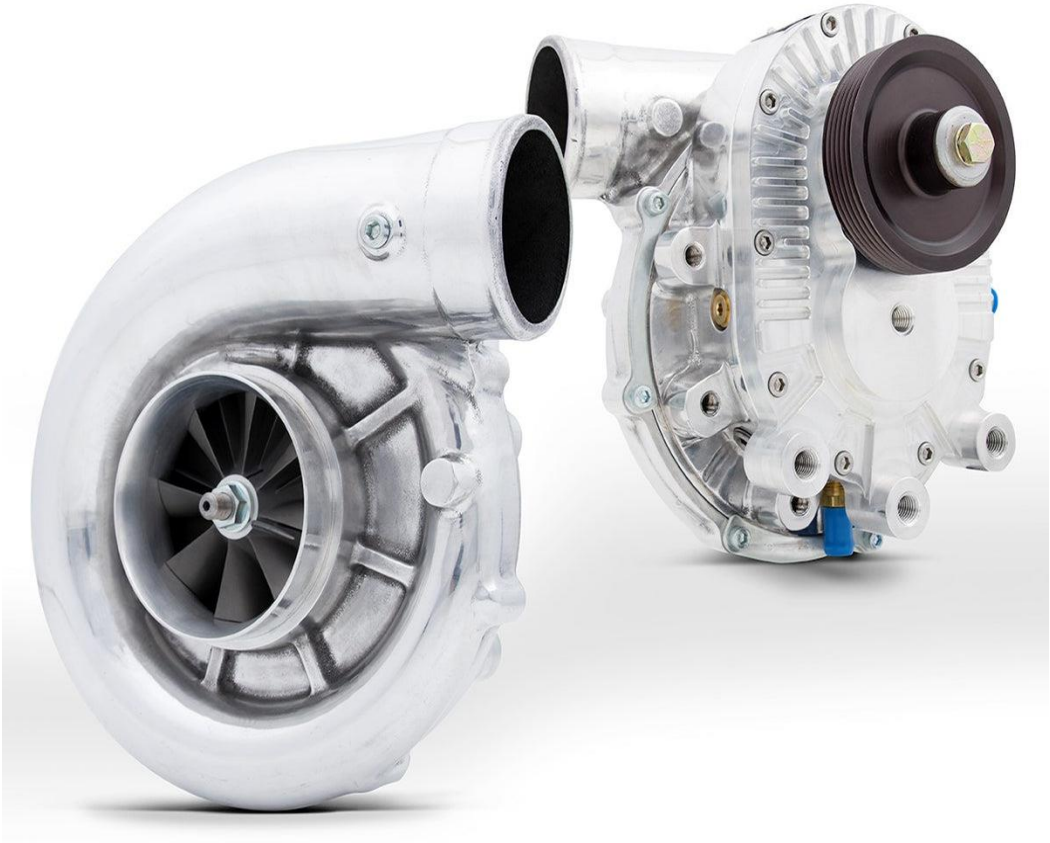
- The points to be noted in supercharging an engine are summarized:
- Supercharging increases the power output of the engine. It does not increase the fuel consumption, per brake kW hour.
- Certain percentage of power is consumed in compressing the air. This power has to be taken from the engine itself. This will lead to some power loss.
- However, it is seen that the net power output will be more than the power output of an engine of the same capacity, without supercharging.



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Supercharging

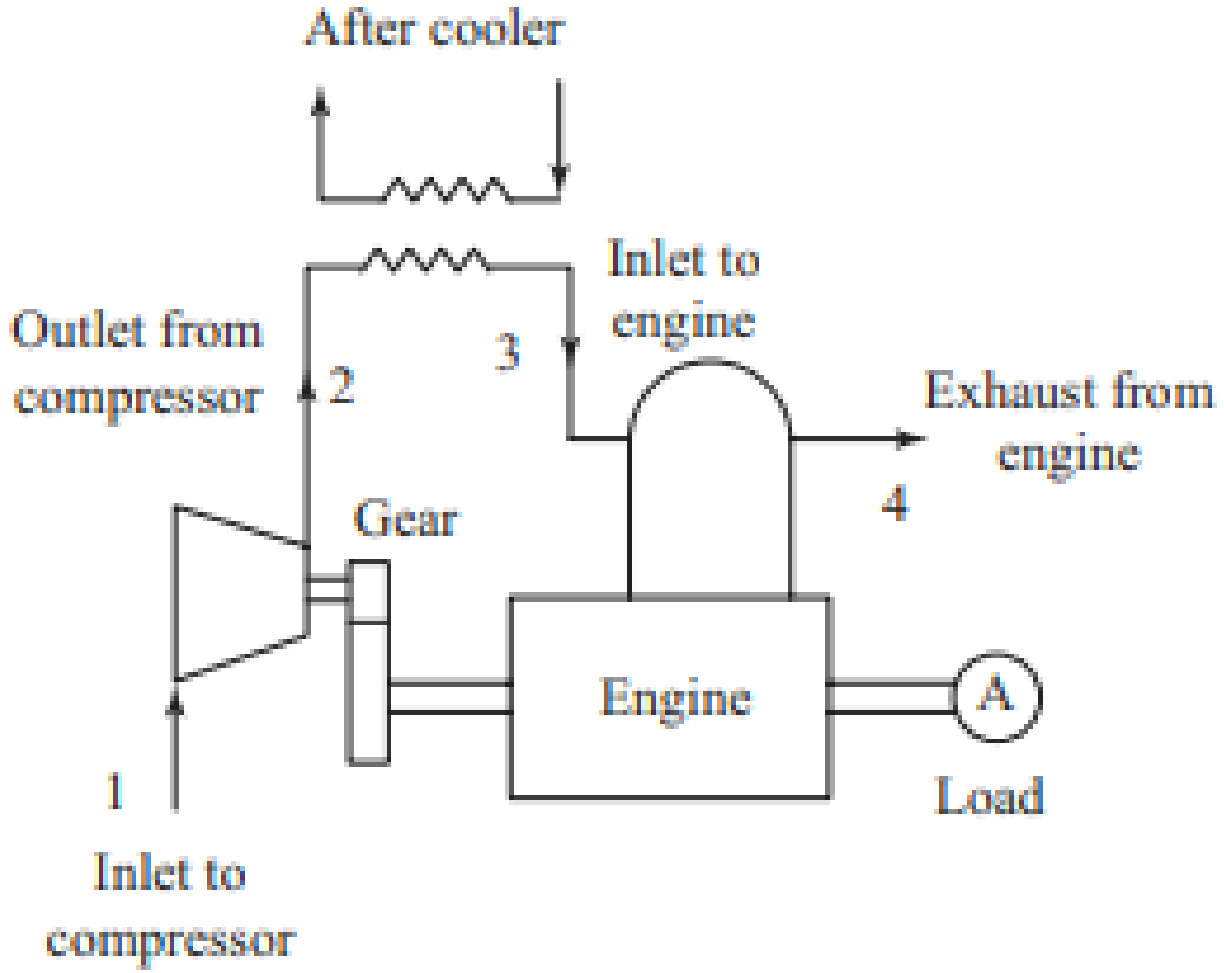
- The engine should be designed to withstand the higher forces due to supercharging.
- The increased pressure and temperature as a result of supercharging, may lead to detonation, therefore the fuel used must have better antiknock characteristics.
- The most important areas where supercharging is of vital importance are :
  - ✓ Marine and automotive engines where weight and space are important.
  - ✓ Engines working at high altitudes. The power loss due to altitude can be compensated by supercharging.



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Supercharging

- In a supercharged marine diesel engine, the air is commonly compressed in a device called a **blower**.
- The blower is driven through gears directly from the engine crankshaft.
- The most common type of blower uses two rotating rotors to compress the air.
- Supercharging is more commonly found on two-stroke engines where the higher pressures that a supercharger is capable of generating are needed.



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Supercharging: Limitations

- Due to supercharging, the thermal load on the various parts of the engine increases. In some engines, the piston head is provided with a hollow space.
- Oil or water is circulated through this space and thereby the piston crown is cooled. In some engines, the piston head and the seat, and the edges of the exhaust valves are usually made of better materials that can withstand higher temperatures.
- Increased gas loading caused by supercharging necessitates the use of larger bearing areas and heavier engine components.
- The permissible extent of supercharging depends upon the ability of the engine to withstand the increased gas loading and thermal stresses.
- Durability, reliability, and fuel economy are the main considerations that limit the degree of supercharging of an engine.
- Because of the increased heat generation and heat transfer, there is a greater tendency to bum the piston head and the seat and edges of the exhaust valves.

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Turbocharging

- In order to meet the regulation of engine emission which is increasingly stringent, turbocharging has become the primary enabler for reducing emissions and boosting fuel economy.
- Nowadays, almost all medium and large diesel engines are equipped with a turbocharger since it increases the mass of air entering the engine to improve both drivability and emissions from engines at the same time.
- However, the applications of turbochargers also lead to higher cylinder back pressure, which may cause more exhaust gases to remain in the cylinder during the exhaust stroke.
- An optimized thermal efficiency can be achieved only when an appropriate turbo compression ratio is selected.
- Turbo charging has played a vital role in the development of the diesel engine.
- **In recent years, the existing research is mainly focused on the performance of turbochargers and their benefits on diesel engines.**

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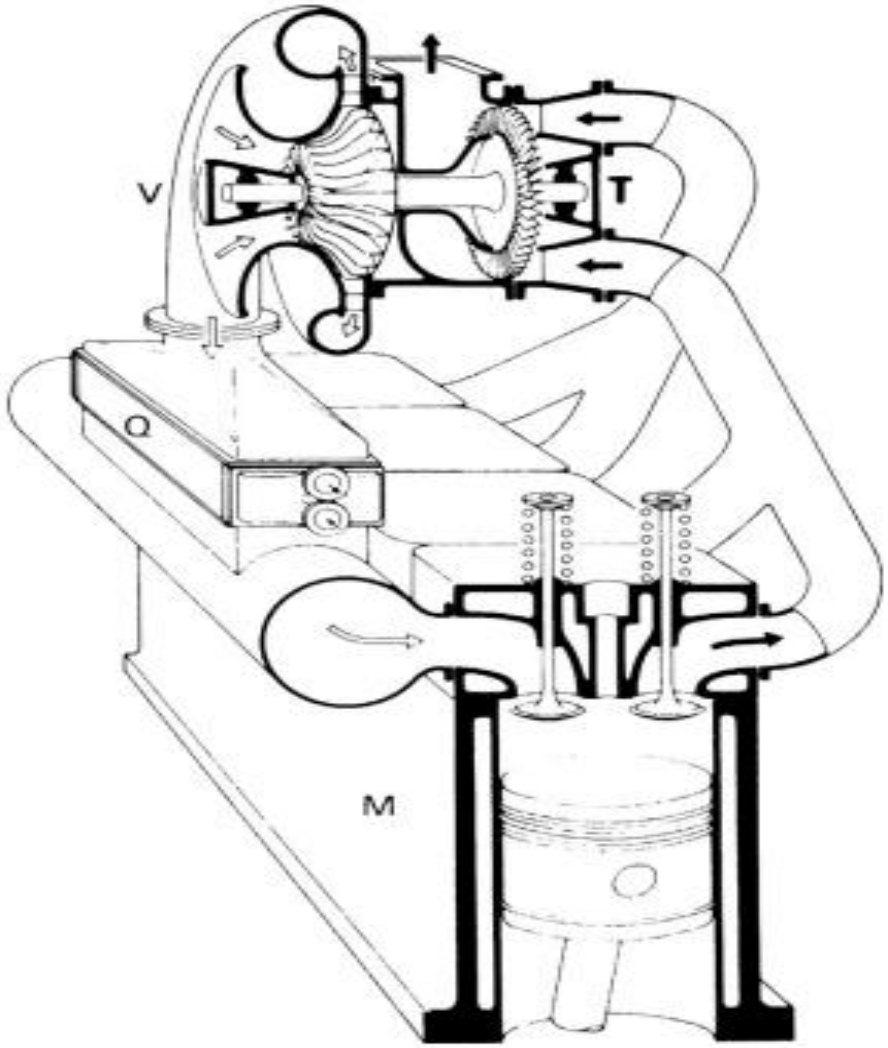
Turbocharging

- A naturally aspirated engine draws air of the same density as the ambient atmosphere.
- This air density determines the maximum weight of fuel that can be effectively burned per working stroke in the cylinder, as well as the maximum power that can be developed by the engine.
- **Increasing the density of the charge air by applying a suitable compressor between the air intake and the cylinder increases the weight of air induced per working stroke, thereby allowing a greater weight of fuel to be burned with a consequent rise in specific power output.**
- The boost in charge air density is accomplished in most modern diesel engine types by exhaust gas turbocharging in which a turbine wheel driven by exhaust gases from the engine is rigidly coupled to a centrifugal-type air compressor. This self-governing process does not require an external governor.

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Turbocharging

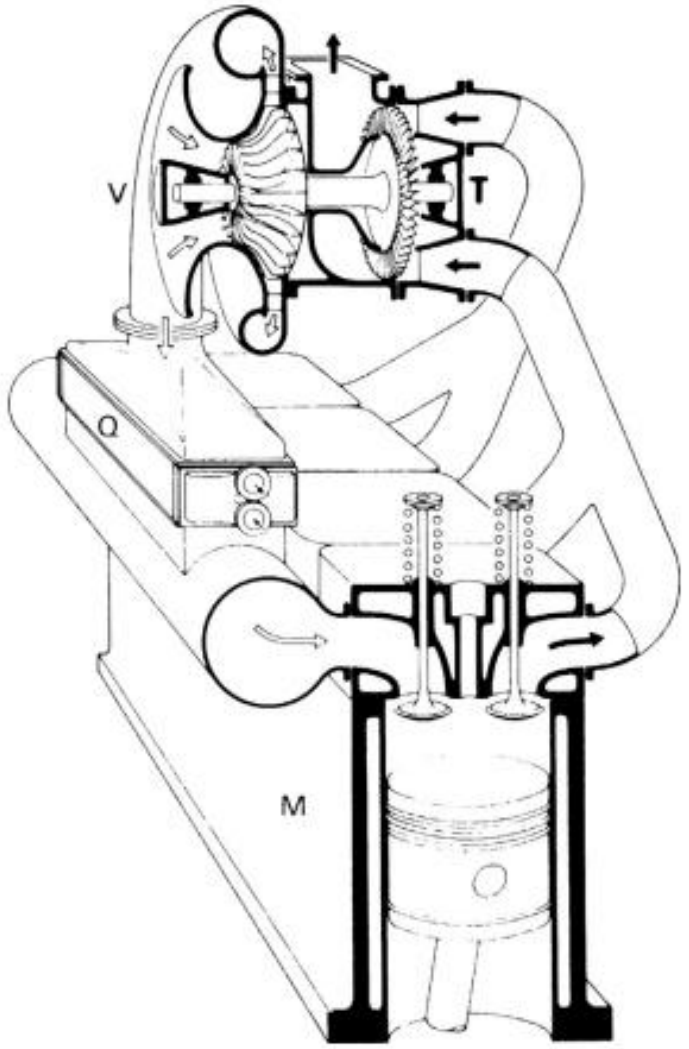
- **A turbocharger consists of a turbine and a compressor on a shared shaft. It converts the heat energy from the exhaust to power, which then drives the compressor to compress ambient air.**
- Normally, the air heated by the compression passes through a cooler which reduces its temperature and increases its density and then is delivered to the air intake manifold of the engine at higher pressure.
- Thus, the amount of air entering the engine cylinders is greater, allowing more fuel to be burnt. As a consequence, the engine produces more power without increasing the engine size.



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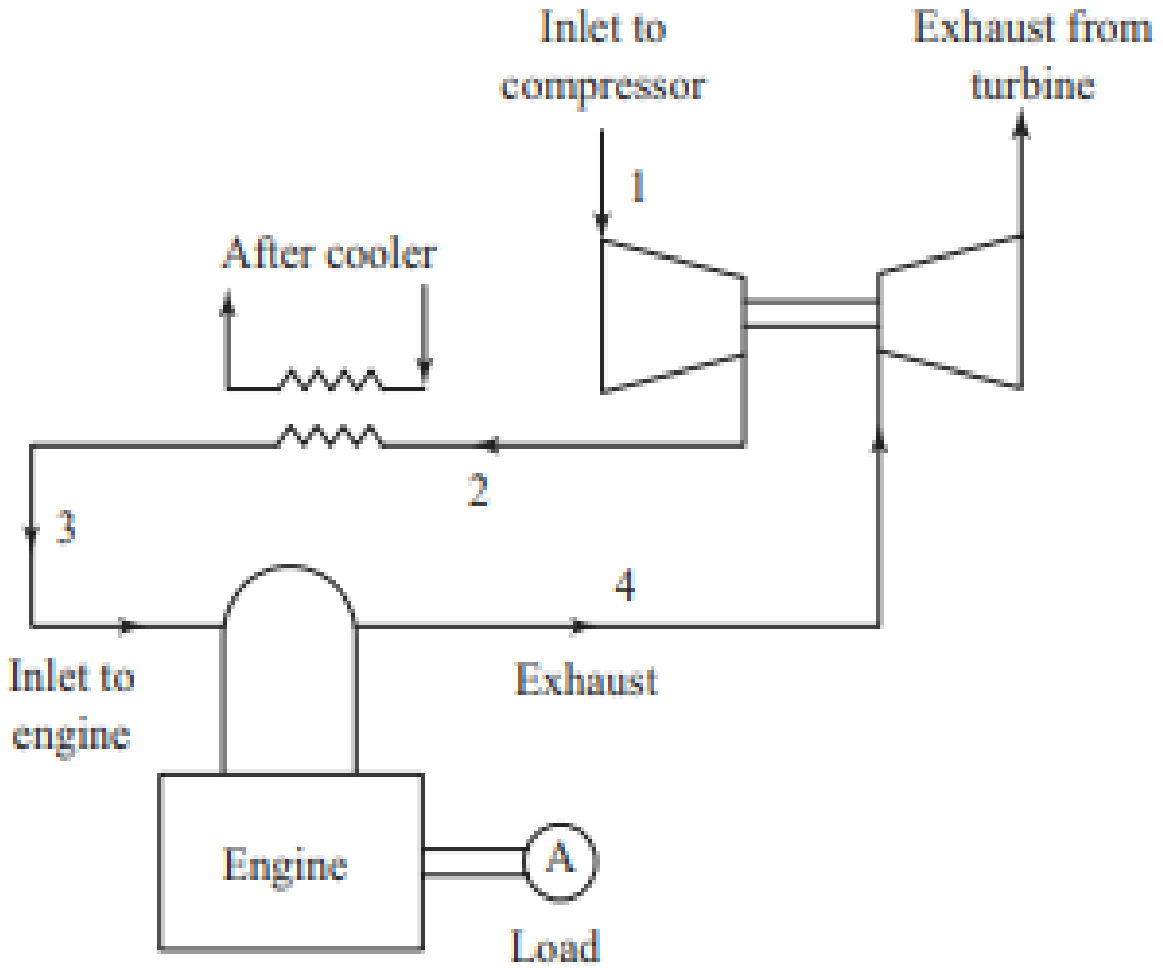
Turbocharging

- There are a number of advantages of pressure charging by means of an exhaust gas turbocharger system;
- ✓ A substantial increase in engine power output for any stated size and piston speed, or conversely a substantial reduction in engine dimensions and weight for any stated horsepower.
- ✓ An appreciable reduction in the specific fuel consumption rate at all engine loads
- ✓ A reduction in initial engine cost
- ✓ Increased reliability and reduced maintenance costs, resulting from less exacting conditions in the cylinders
- ✓ Cleaner emissions
- ✓ Enhanced engine operating flexibility



Turbocharging

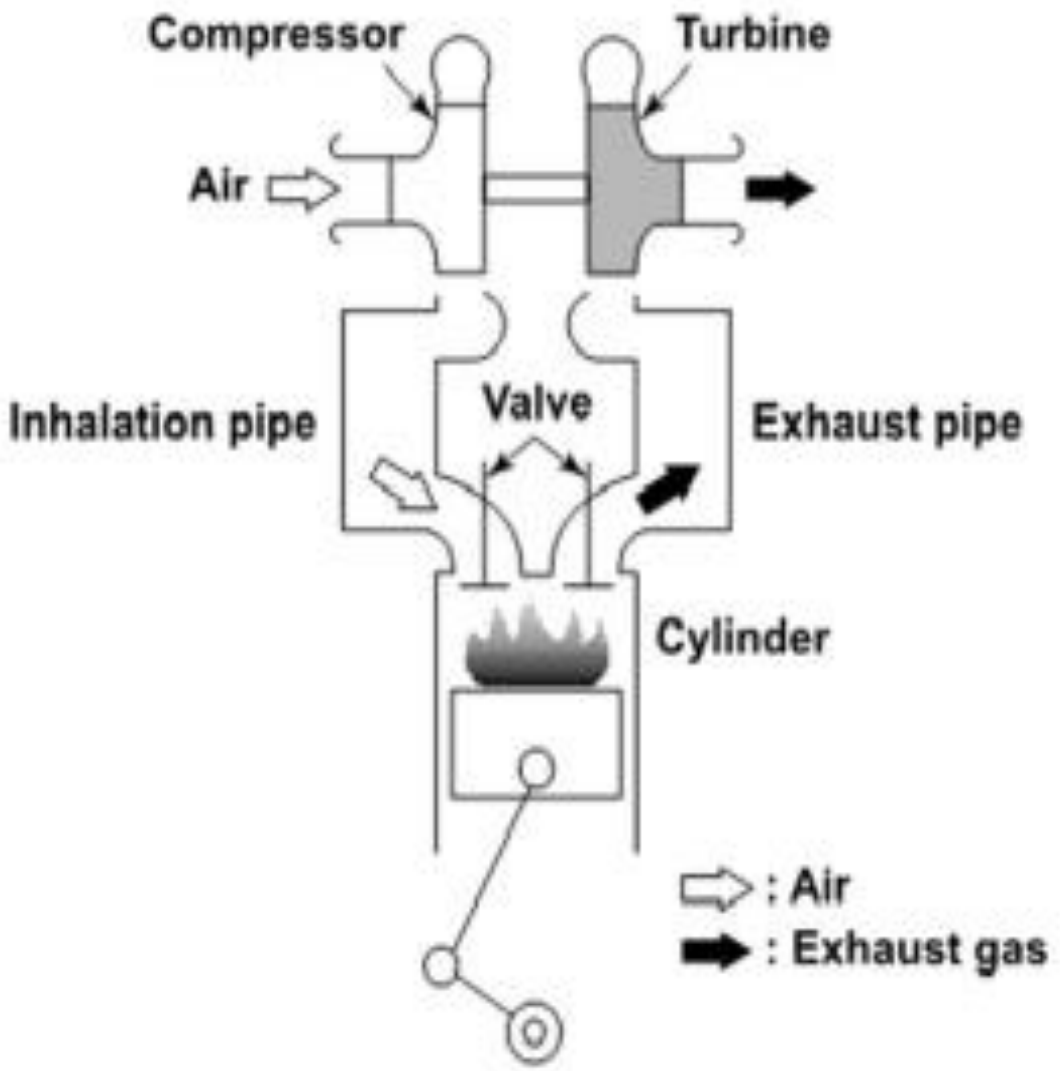
- The power expended in driving the compressor has an important influence on the operating efficiency of the engine.
- About 35% of the total heat energy in the fuel is wasted on the exhaust gases, so by using the energy in these gases to drive the compressor, an increase in power is obtained in proportion to the increase in the charge air density.



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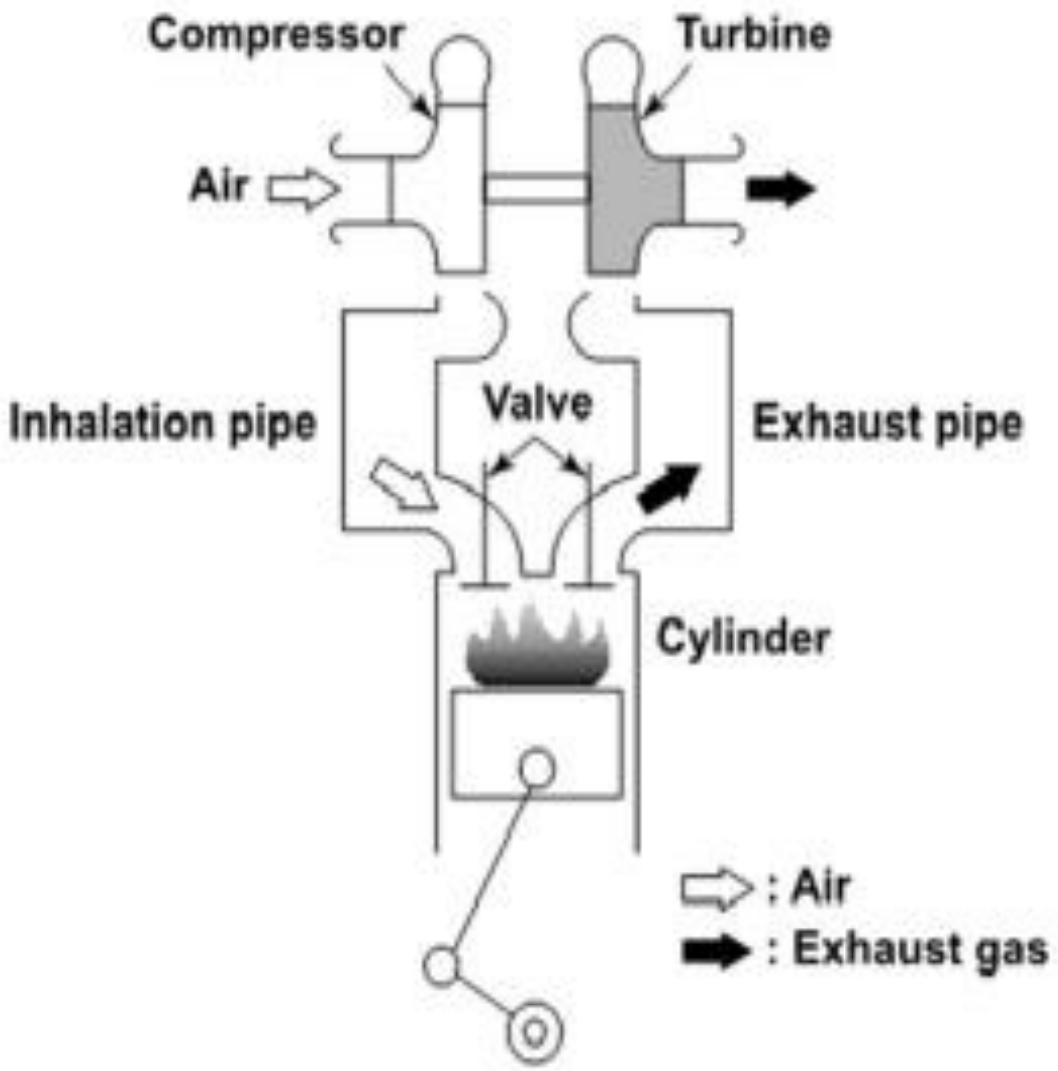
Turbocharging: Four-stroke engines

- Exhaust gas turbocharged four-stroke marine engines can deliver three times or more power than naturally aspirated engines of the same speed and dimensions.
- In matching the turbo blower to the engine, a free air quantity in excess of the swept volume is required to allow for the increased density of the charge air and to provide sufficient air for through-scavenging of the cylinders after combustion.
- In a typical case, to ensure adequate scavenging and cooling of the cylinders, the air inlet valve opens at 80° bTDC and closes at 40° aBDC. The exhaust valve opens at 50° before BDC and closes at 60° after TDC.



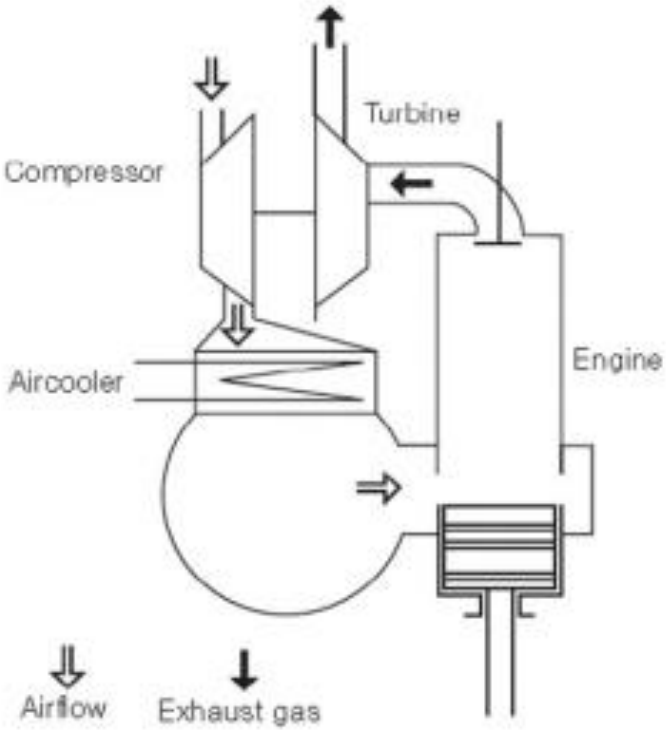
Turbocharging: Four-stroke engines

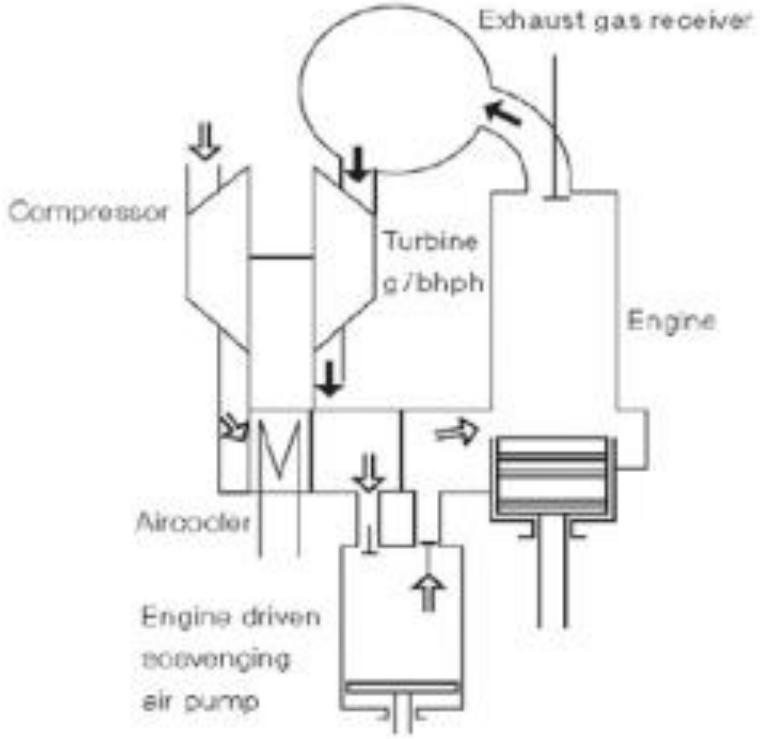
- The engine exhaust system should be so designed that gases from one cylinder can't contaminate the charge air in another cylinder, either by blowing back through the exhaust valve or by interfering with the discharge of gases from the cylinder.
- During the period of valve overlap, it follows that the exhaust pressure must be less than the air charging pressure to ensure effective scavenging of the cylinder to remove residual gases and cooling purposes.



Turbocharging: Two-stroke engines

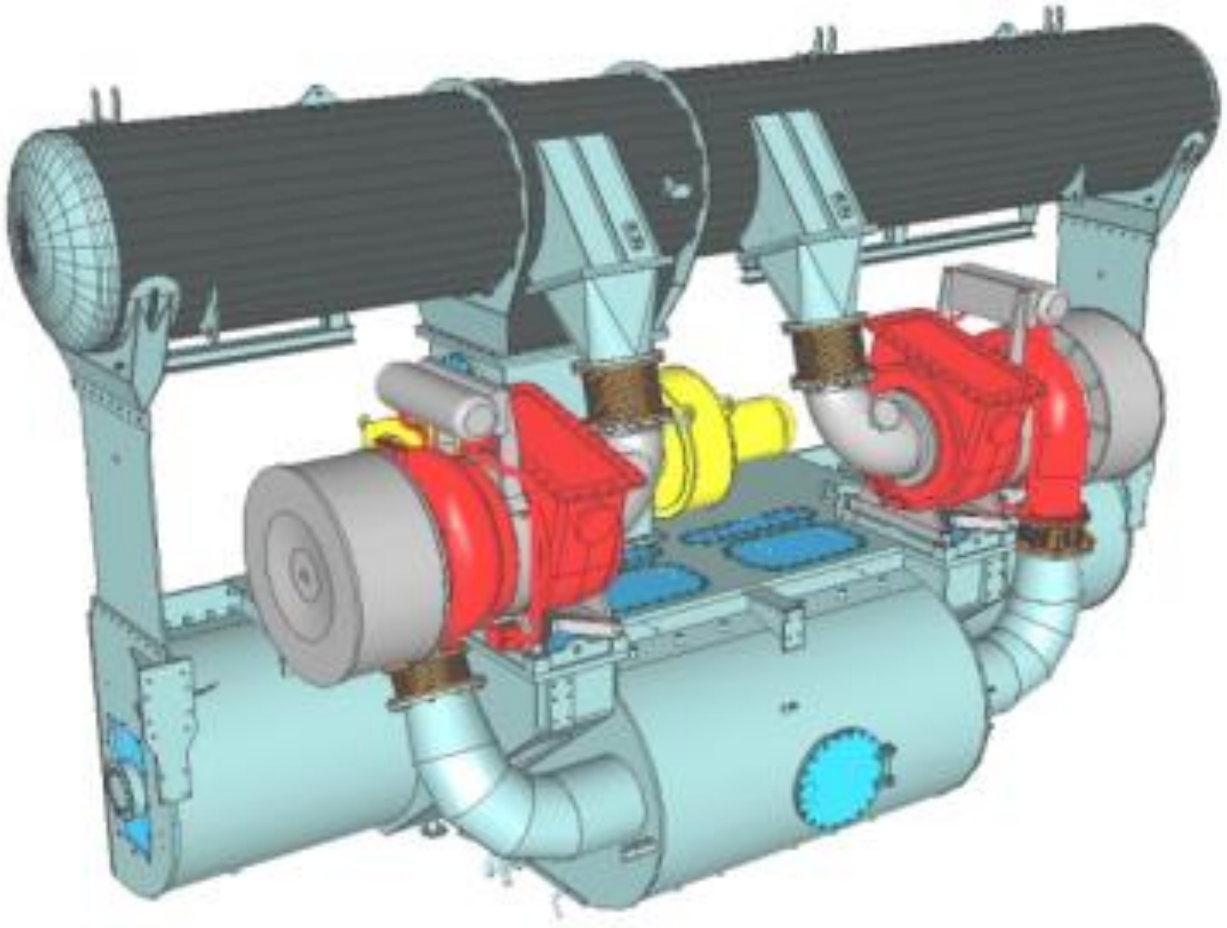
- Turbocharging and scavenging arrangements for two-stroke engines are now fairly standardized.
- All such engines are today uniflow scavenged, with air entering the cylinder through ports around the full circumference of the cylinder liner at the bottom of the piston stroke and exhausting via a single poppet-type valve in the centre of the cylinder cover.





Turbocharging: Two-stroke engines

- The turbochargers operate on a constant pressure system, with scavenging air pressures up to 3.9 bar.
- There are one to four turbochargers, all located high on the side of the engine, outboard, and beneath the exhaust manifold.
- This allows an efficient arrangement with the scavenge air cooler and its associated water separator unit located immediately below the turbocharger and adjacent to the scavenge air space on the piston underside.

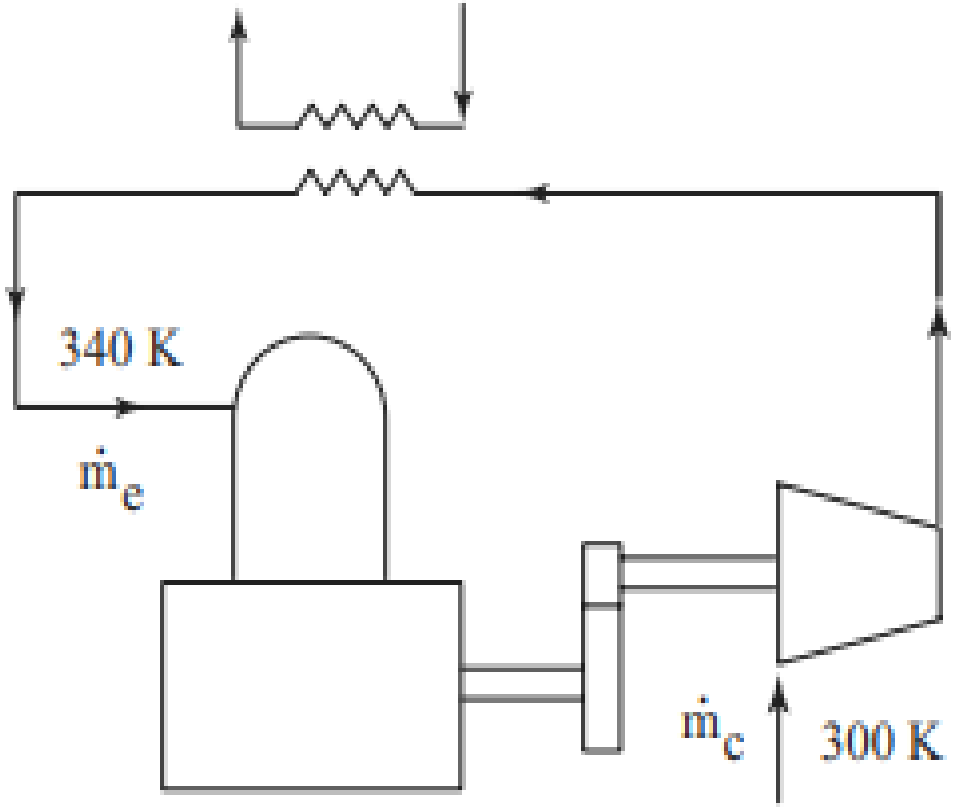


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Supercharging: Problem

- A mechanically coupled supercharger is run by a four-stroke four-cylinder square diesel engine with a bore of 120 mm as per the arrangement shown.
- Air enters the compressor at 27 °C and the compressor pressure ratio is 1.6. From the compressor, the air is passed on to a cooler where 1200 kJ/min of heat is rejected. The air leaves the cooler at 67 °C before entering the engine.
- The volumetric efficiency is 85% based on intake manifold pressure and temperature. The other details about the engine are: bp is 50 kW, speed is 3000 rpm, and mechanical efficiency η_m is 85%. Take the isentropic efficiency of the compressor to be 85%, the density of air to be 1.64 kg/m^3, cp to be 1.005 kJ/kg.K, and γ to be 1.4. Determine;

✓ imep of the engine



✓ air consumption rate of the engine

✓ the air handling capacity of the compressor in kg/min.

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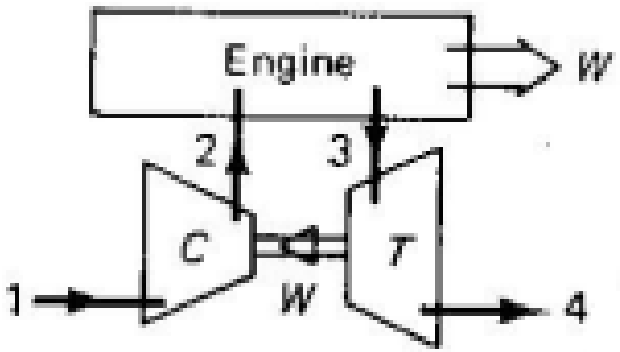
Turbocharging: Problem

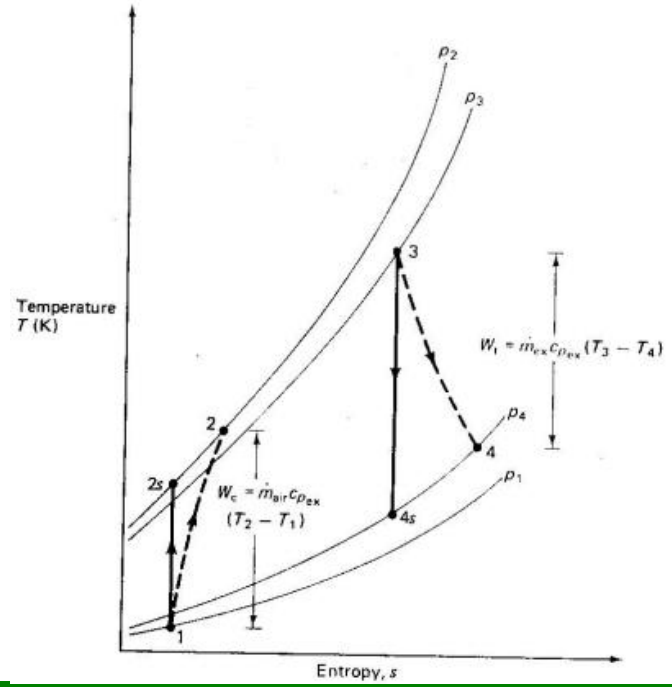
- A diesel engine is fitted with a turbocharger, which comprises a radial compressor driven by a radial exhaust gas turbine. The air is drawn into the compressor at a pressure of 0.95 bar and at a temperature of 15°C and is delivered to the engine at a pressure of 2.0 bar.
- The engine is operating with an air/fuel ratio of 18:1 by mass, and the exhaust leaves the engine at a temperature of 600 °C and at a pressure of 1.8 bar; the turbine exhausts at 1.05 bar. The isentropic efficiencies of the compressor and turbine are 70 percent and 80 percent, respectively.
- Using the values; c_{pair} = 1.01 kJ/kg K, γ_{air} = 1.4 and c_{pex} = 1.15 kJ/kg K, γ_{ex} = 1.33. Calculate

✓ The temperature of the air leaving the compressor

✓ The temperature of the gases leaving the turbine

✓ The mechanical power loss in the turbocharger expressed as a percentage of the power generated in the turbine.






THANK
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